Electronic Design Project

Car Dashboard

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Objective

Design an Arithmetic and Logic Unit (ALU) capable of performing a set of arithmetic and logical operations on 4-bit operands.

Selected Operations and Their Roles in the Project

Operation / Component	Role in the Project
Multiplication	Calculate the remaining distance to travel (in miles).
Synchronous Random-Cycle Counter	Test the car's sensors at startup.
Asynchronous Binary Counter	Count the engine's rotational speed (in miles per hour).
Left/Right Shift Register	Animate the turn signal lights.
Linear Feedback Shift Register (LFSR)	Generate a changing security code for the keyless entry system.
Encoder / Decoder	Translate the gear lever's physical position to display the corresponding mode (p, r, n, d).
Moore Finite State Machine (FSM)	Control the windshield wipers according to rain conditions (Off, Continuous, Fast).

Display of Results

All results and the states of the different components will be displayed using the LEDs and 7-segment displays on the electronic development board.